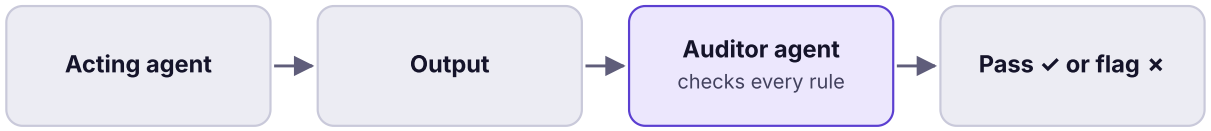
Explainability & Compliance Patterns

1. Staying on the rules

- **Instruction Fidelity Auditing**: a separate auditor checks the output (and reasoning) against every stated instruction and flags any skipped or violated. Example: confirming an e-commerce agent applied exactly the eligible discounts
- Why a separate auditor? An agent **rationalises its own mistakes**; objective checking needs an external evaluator
- **Persistent Instruction Anchoring**: re-inject the non-negotiable constraints at every step of a long run so they stay salient (fights lost-in-the-middle). Anchor the critical constraints only, because it costs context

✎ DIAGRAM



The agent that acts shouldn't be the one that audits.

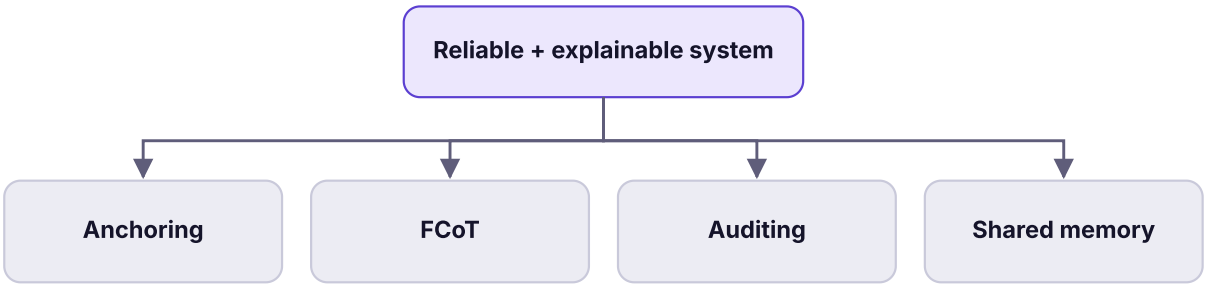
2. Making reasoning inspectable

- **Fractal Chain-of-Thought (FCoT)**: structured reasoning embedded at every level (orchestrator and each specialist), composing into one auditable chain, so a reviewer can trace why the system concluded something at any depth
- **Shared Epistemic Memory**: a common, versioned knowledge store all agents read and write, so they act from one consistent picture of the world (e.g. a supply-chain disruption response)

3. Compose for systemic reliability

- Anchor the constraints + embed FCoT for transparency + audit for fidelity + share epistemic memory for consistency → a system that is both reliable and explainable (maturity Level 4)
- In the loan use case, FCoT guides the agent's reasoning so every approve or reject decision comes with legible, auditable reasoning

✎ DIAGRAM



Explainability is designed into how the agent reasons, not added as a report later.

SAY IT

"Explainability isn't a report you add at the end; it's designed into how the agents reason, with a separate auditor checking the output against every instruction."